

Chapter 1: The Linux Foundation

LFS101x — Introduction to Linux | Easy Notes for Revision

Introduction

This chapter introduces who The Linux Foundation is, why they created this course, and what you need (hardware and mindset) before starting the hands-on parts of the course. It's short, but it sets up important context you'll refer back to later.

What Is The Linux Foundation?

The Linux Foundation is a non-profit organization founded in 2000. Its job is to provide a neutral, trusted home where developers, companies, and communities can build and grow open source software together.

In simple words:

- It doesn't own Linux — it supports the people and companies who build it.
- It's "neutral," meaning no single company controls decisions.
- It's backed by 1,000+ member companies (including giants like IBM, Google, Microsoft).

What Does The Linux Foundation Actually Do?

- **Hosts open source projects** — Linux itself is the most famous one, but they host hundreds of others (Kubernetes, Node.js, Hyperledger, etc.) across cloud, networking, security, and embedded systems.
- **Runs events** — conferences like Open Source Summit and KubeCon bring together 85,000+ developers and companies every year to share knowledge.
- **Provides training** — classroom, online, on-site, and event-based courses (this course, LFS101x, is one of them).
- **Offers certifications** — performance-based or multiple-choice exams that prove your skills to employers, which you can take from anywhere, anytime.

Why This Matters For You

The Linux Foundation isn't teaching you Linux to promote one company's product — that's the whole point of being "vendor-neutral." This means the skills you learn apply across Ubuntu, Fedora, CentOS, and every other distribution, not just one.

Course Linux Requirements

Since this is a hands-on course, you need an actual Linux system to practice on. The course doesn't force you to pick one distribution — it teaches concepts that apply broadly, using three representative distros:

- **Red Hat family** — Fedora, RHEL, CentOS
- **Debian family** — Debian, **Ubuntu**, Linux Mint
- **SUSE family** — SUSE, openSUSE

Beginners are usually advised to start with **Ubuntu** or **Linux Mint**, since they're the most beginner-friendly.

Three Ways to Get Linux Running

- **Live CD/USB** — run Linux temporarily without installing anything. Safe, but slower performance and changes aren't saved.
- **Virtual Machine (recommended for beginners)** — install a hypervisor (like Oracle VirtualBox) on your existing OS, then run Linux inside it in a window. Your main computer stays completely untouched.
- **Native Install** — install Linux directly on your hard drive, either replacing your current OS or dual-booting alongside it. Riskier, involves disk partitioning.

My Setup (for reference)

I'm using the Virtual Machine method — Oracle VirtualBox as the hypervisor, running Ubuntu inside it. Recommended minimums: at least 20 GB disk space and 2 GB+ RAM assigned to the VM. This also requires hardware virtualization (Intel VT-x / AMD-V) to be enabled in the computer's BIOS/UEFI settings.

Why This Matters for Cloud Computing & DevOps

You're not just learning Linux for its own sake — here's the direct line from this chapter to your career goals:

- **Cloud servers run Linux.** AWS, Azure, and Google Cloud all offer Linux virtual machines as their most common option — most cloud workloads worldwide run on Linux, not Windows.
- **DevOps tools are built for Linux first.** Docker, Kubernetes, Jenkins, Terraform, and Ansible were all designed to run natively on Linux — you'll be typing Linux commands constantly in DevOps work.
- **Virtual Machines are a DevOps skill too.** The VirtualBox + Ubuntu setup you just did is the same basic idea behind cloud VMs (like an AWS EC2 instance) — you're literally practicing a real DevOps concept right now.
- **Vendor-neutral skills = flexible career.** Because The Linux Foundation doesn't push one distro, what you learn transfers to any company's Linux setup, whether they use Ubuntu, RHEL, or Amazon Linux.
- **Certifications are recognized by employers.** Linux Foundation certifications (like LFCS, KCNA) are widely respected in cloud/DevOps hiring — this course is the on-ramp to those.

Chapter Summary

- The Linux Foundation is a neutral non-profit that supports open source projects, not just Linux.
- It provides training, certifications, and hosts major industry events.
- This course uses three distro families as examples: Red Hat, Debian (Ubuntu), and SUSE.
- You need a working Linux environment to follow along — Live media, Virtual Machine, or Native install.
- A Virtual Machine is the safest and most recommended method for beginners.

Beginner-Friendly Terms

Open Source

Software whose source code is publicly available so anyone can view, modify, or contribute to it.

Non-profit

An organization that reinvests any income back into its mission instead of distributing profit to owners.

Distribution (Distro)

A complete, ready-to-use Linux operating system built around the Linux kernel plus tools, apps, and a package manager.

Hypervisor

Software that lets you run one operating system (like Ubuntu) inside a "virtual computer" on top of another (like Windows).

Virtual Machine (VM)

A simulated computer running inside your real computer, created by a hypervisor. Safe to experiment with — it can't damage your actual system.

BIOS/UEFI

Low-level firmware that starts your computer before the operating system loads. Settings like virtualization support live here.

VT-x (Intel) / AMD-V

A hardware feature in modern CPUs that must be enabled in BIOS for virtual machines to run properly.

Certification

A formal exam-based credential from The Linux Foundation that proves your Linux skills to employers.

Member Companies

Businesses (like IBM, Google, Microsoft, Samsung) that pay to fund and support The Linux Foundation's work, without controlling its decisions.

Kernel

The core program of an operating system that talks directly to the computer's hardware (CPU, memory, disk). Linux itself, strictly speaking, refers to just this kernel — everything else (apps, desktop) is added by a distribution.

Package Manager

A tool built into a Linux distribution used to install, update, and remove software easily (e.g. apt on Ubuntu, dnf on Fedora).

Cloud Computing

Renting computing power (servers, storage) over the internet from a provider (AWS, Azure, GCP) instead of buying and running your own physical hardware.

DevOps

A way of working that combines software Development and IT Operations, using automation to build, test, and deploy software faster and more reliably.

Native Install

Installing an operating system directly onto a computer's real hard drive, replacing or sharing space with the existing OS — as opposed to running it inside a VM.

Dual-boot

Having two operating systems (e.g. Windows and Linux) installed on the same physical computer, letting you choose which one to start each time you power on.

Quick Revision

- The Linux Foundation was founded in 2000 and is vendor-neutral.
- It hosts Linux plus hundreds of other open source projects.
- It offers training, certifications, and events (like KubeCon, Open Source Summit).
- Three distro families used in this course: Red Hat, Debian, SUSE.
- Beginners should start with Ubuntu or Linux Mint.
- Three install methods: Live media, Virtual Machine, Native install.
- Virtual Machine = safest way to learn Linux without risking your main OS.
- Enabling VT-x/AMD-V in BIOS is required before a VM can run.

Interview Questions

Q1. What is The Linux Foundation?

A non-profit organization founded in 2000 that provides a neutral home for open source collaboration, and hosts Linux along with many other open source projects.

Q2. Does The Linux Foundation own Linux?

No. It supports the community and infrastructure around Linux, but Linux itself is developed openly by contributors worldwide, led by Linus Torvalds and maintainer Greg Kroah-Hartman.

Q3. Name the three Linux distribution families covered in this course.

Red Hat (Fedora, RHEL, CentOS), Debian (Debian, Ubuntu, Linux Mint), and SUSE (SUSE, openSUSE).

Q4. What are the three ways to get Linux running on a computer?

Live CD/USB (no install), Virtual Machine (install a hypervisor), and Native Install (directly on the hard disk).

Q5. Why is a Virtual Machine recommended for beginners?

Because it runs as a regular application on top of your existing OS — if something goes wrong inside the VM, your main computer and files stay completely safe.

Q6. What is VT-x and why does it matter?

VT-x is Intel's hardware virtualization feature (AMD-V for AMD chips). It must be enabled in BIOS/UEFI before software like VirtualBox can create and run virtual machines.

Q7. What kind of exams does The Linux Foundation certification program offer?

Performance-based and multiple-choice exams, distribution-flexible, that can be taken remotely from anywhere.

Personal note: keep this concise — the goal is to be able to revise this chapter in 5–10 minutes, not to memorize every detail from the course.